

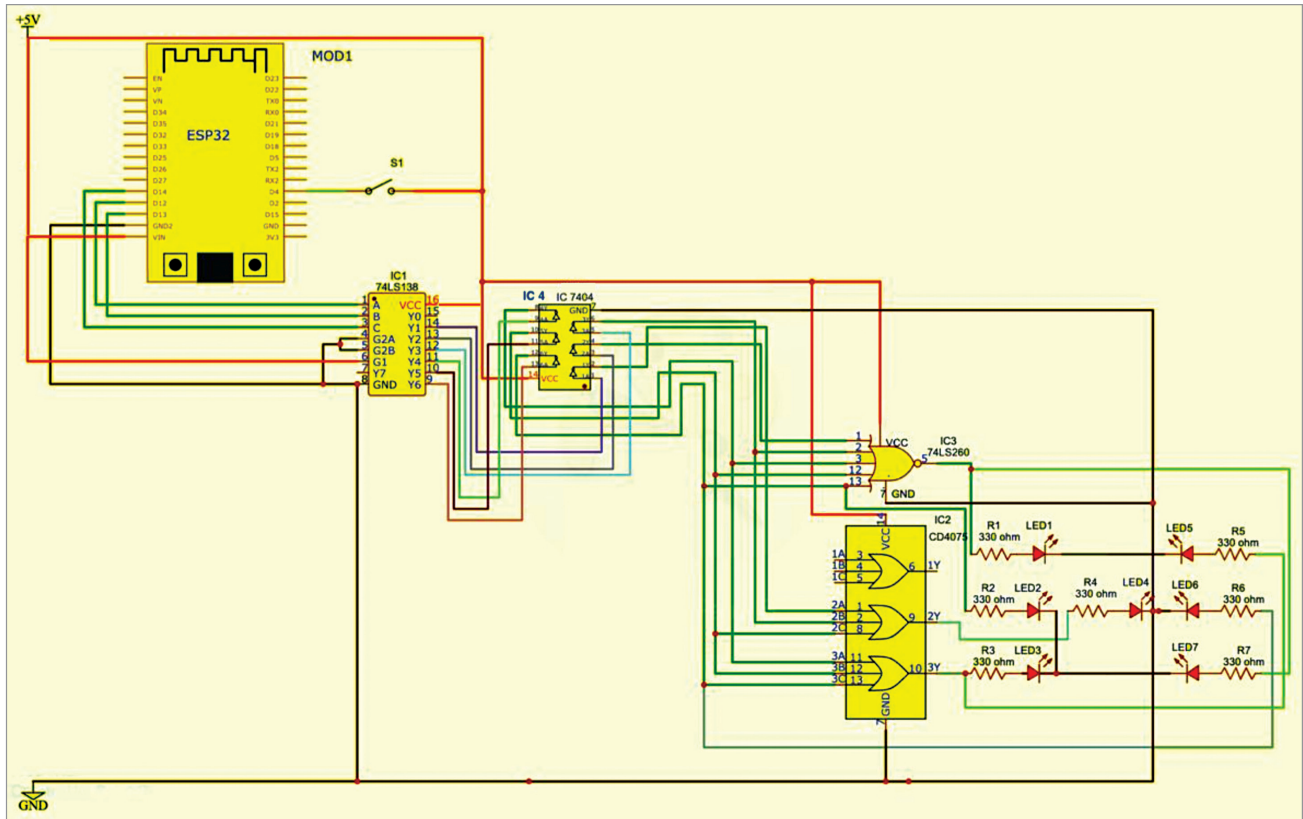


Fig. 5: Circuit diagram of the proposed system

to logic high.

The C bit is the MSB, and the A bit is the LSB, as shown in Table 2.

With or without a Karnaugh map, the expressions for each valid dice face are derived and implemented using AOI logic. The modification requires three additional ICs—74LS04, 74LS260, and

CD4075—to provide the correct logic and generate seven outputs. These outputs are linked to seven LEDs, representing the dice faces.

Each decoder output line is complemented to counteract the negative logic. The final output is displayed in a pattern that simulates the face of an actual dice. The

rolling effect lasts approximately five to six seconds—four seconds for rolling and one to two seconds for decoding and display.

Fig. 3 illustrates the timing diagram of the e-dice throw, and Fig. 4 shows the LED arrangement, ensuring accurate representation of all possible dice outcomes.

Circuit and working

Fig. 5 shows the circuit diagram of the e-dice, built around:

- ESP32 board
 - 3-to-8 decoder (IC74LS138 – IC1)
 - Three-input OR gate (4075 – IC2)
 - Five-input NOR gate (74LS260 – IC3)
 - Hex inverter (7404 – IC4)
 - Other supporting components
- The code is developed using Arduino IDE. In the code:
- Input and output pins for the dice are defined.

TABLE 2

TRUTH TABLE OF THE 3-TO-8 DECODER 74LS138

Input						Output							
G1	G2A	G2B	C (MSB)	B	A (LSB)	Y7	Y6	Y5	Y4	Y3	Y2	Y1	Y0
1	0	0	0	0	0	1	1	1	1	1	1	1	0
1	0	0	0	0	1	1	1	1	1	1	1	0	1
1	0	0	0	1	0	1	1	1	1	1	0	1	1
1	0	0	0	1	1	1	1	1	1	0	1	1	1
1	0	0	1	0	0	1	1	1	0	1	1	1	1
1	0	0	1	0	1	1	1	0	1	1	1	1	1
1	0	0	1	1	0	1	0	1	1	1	1	1	1
1	0	0	1	1	1	0	1	1	1	1	1	1	1

Note: Outputs Y0 and Y7 of 74LS138 are not used in circuit